



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

APACHE CASSANDRA PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

Q1 What type of database is Apache Cassandra and what are its core strengths? [Architecture](#)

Q6 How do you monitor and tune slow queries in Apache Cassandra? [Observability](#)

Q2 How does Apache Cassandra guarantee consistency and handle transactions? [Architecture](#)

Q7 What backup and restore strategies does Apache Cassandra support? [Lifecycle](#)

Q3 What index types does Apache Cassandra support and when do you use each? [Indexing](#)

Q8 How do you handle schema migrations in Apache Cassandra? [Schema](#)

Q4 How do you design a schema or data model in Apache Cassandra? [Hardening](#)

Q9 What security features does Apache Cassandra provide? [Testing](#)

Q5 How does Apache Cassandra scale horizontally? [SSO / IdP](#)

Q10 What are common anti-patterns when using Apache Cassandra in production? [Performance](#)



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Apache Cassandra is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

Mitigation

A6 Apache Cassandra provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

Observability

A2 Apache Cassandra provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

Limitations

A7 Apache Cassandra supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

Information

A3 Apache Cassandra offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

Hardening

A8 Schema migrations in Apache Cassandra are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

Governance

A4 Schema design in Apache Cassandra involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

Optimization

A9 Apache Cassandra supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

Assessment

A5 Apache Cassandra scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some Apache Cassandra implementations handle this automatically; others require manual configuration.

Consideration

A10 Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.

Optimization